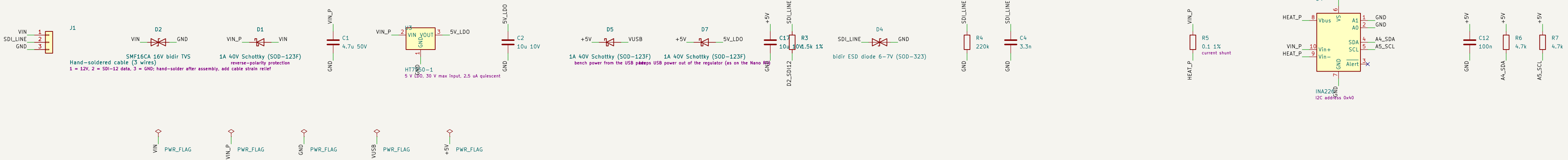


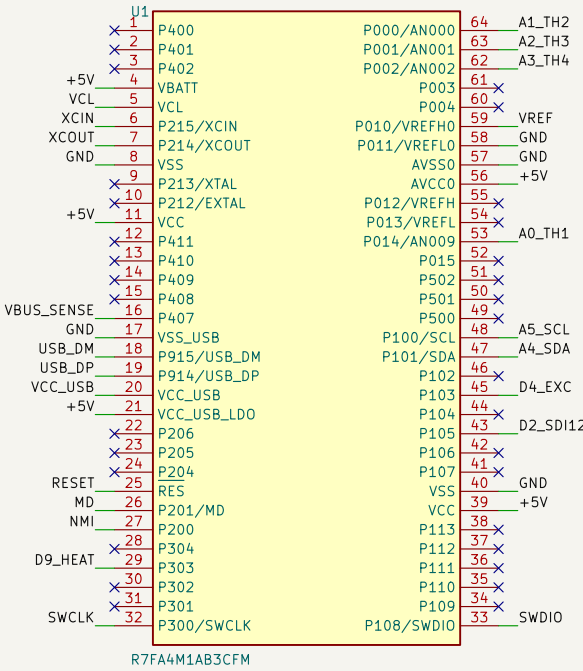
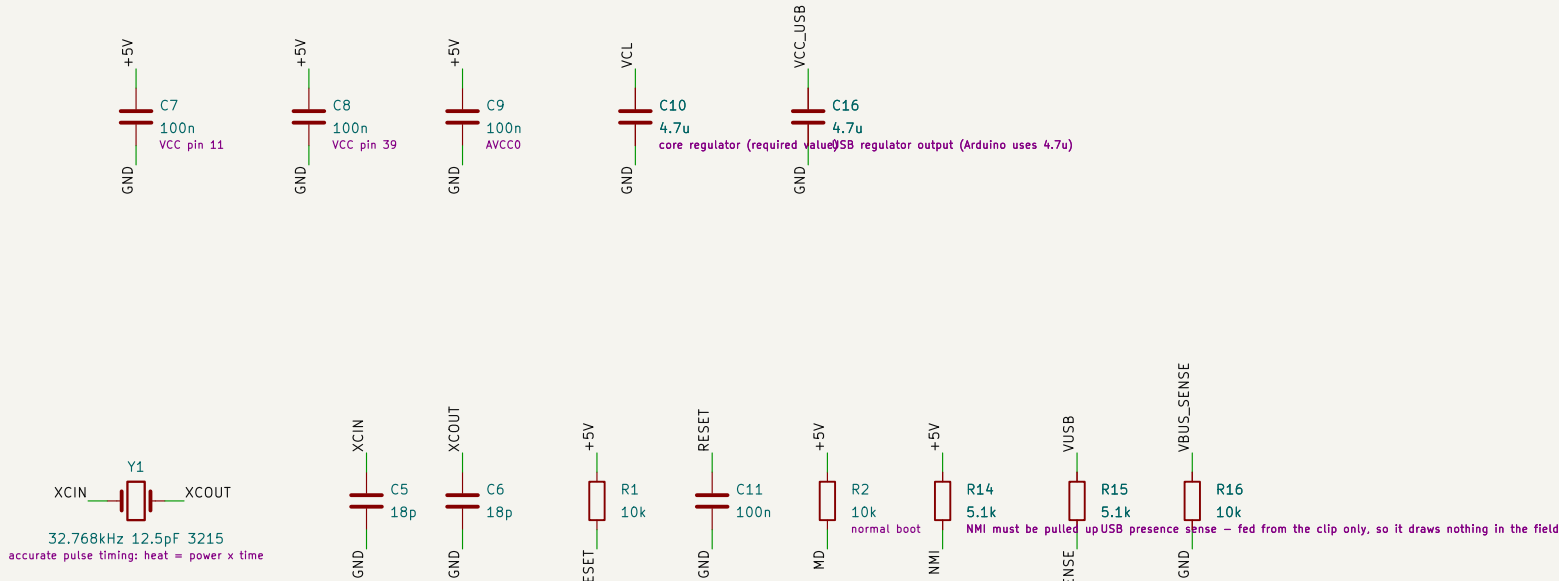
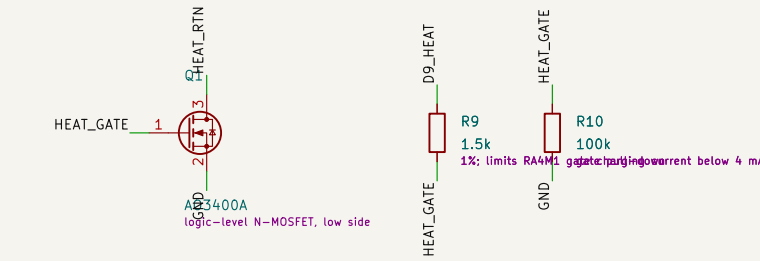
HP-SDI12-NANO r2 / COMPACT FLAT – 11.5–14.4 V battery supply over the 3-wire SDI-12 cable. J1 has three bare cable solder holes (VIN, data, ground); no connector is fitted. Asleep the board draws only the regulator (2.5 uA) plus MCU standby current.

SDI-12 DATA – meets the SDI-12 electrical numbers: R3 sets the 1–2 k transmit resistance (and limits fault current into pin D2), R4 the 160–360 k idle resistance, C4 holds the edge rate under 1.5 V/us. D4 clamps cable transients.

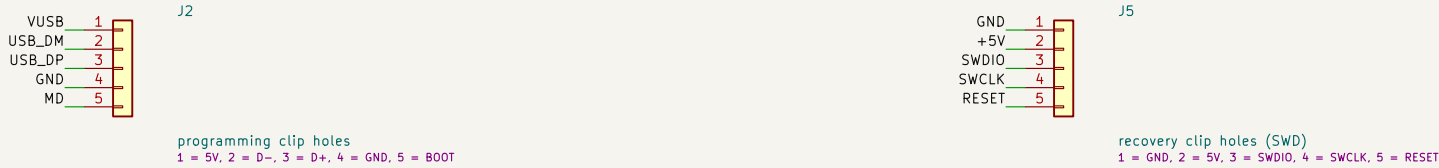
HEATER DRIVE + POWER MONITOR – PWM-capable D9/P303 drives Q1. Required firmware: 100 Hz, nominal 2 W average, 8–15 s; integrate stable ON power times actual ON time. Force D9 LOW at shutdown; stopping a timer is insufficient. R10 pulls an undriven gate LOW. Keep all three INA226 inputs on their dedicated Kelvin connections. Firmware is not included.



MICROCONTROLLER – Renesas RA4M1 (R7FA4M1AB3CFM), the Arduino UNO R4 / Nano R4 chip, wired exactly like the UNO R4 Minima so the stock board package and Arduino pin names apply (nets D2, D4, D9, A0–A5). Runs at 5 V. Support circuit (NMI, USB sense, VCC_USB, VCL) follows Arduino's Nano R4 schematic.



FIRMWARE ACCESS (no soldering) – J2 is a row of five holes for an off-the-shelf 2.54 mm 5-pin pogo clip. Holes 1–4 carry USB (5V, D–, D+, GND); plug a 'USB to 4-pin Dupont' cable onto the clip and the board is an 'Arduino UNO R4 Minima' in the IDE. Hole 5 = BOOT: only for the very first load of the Arduino bootloader, put a jumper cap across clip pins 4–5 while plugging in. J5 = a second five-hole row (GND, 5V, SWDIO, SWCLK, RESET) for a debug probe if a board ever needs recovery.



HEATER RESISTOR CHAIN (top of centre prong) – 17 x 3.3 ohm 1% 0603 in series, 2.6 mm pitch. EXACT Vishay CRCW0603R30FKEAHP, 0.33 W at 70 C; no ordinary 0.1 W substitutes. Return is on bottom copper. Nominal full-ON power 2.34 W at 12 V, 3.41 W at 14.4 V with specified cable losses. PWM controls average energy; instantaneous ratings still apply. Qualify the potted needle thermally.

